

PHASE 7 → PHASE 10

AI-Driven Model Engineering SLR — Simple Group Work Guide for High Marks

Use this as your group's action plan. Phase 7 is Analysis, Phase 8 is Synthesis, Phase 9 is your scholarly contribution, and Phase 10 is the final report. The rubric makes it clear that high marks require critical synthesis, comparison, evidence weighting, original evidence-based artefacts, and professional presentation — not just summaries.

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THE BIG PICTURE

Phase 7: Look at the papers together and find patterns.

Phase 8: Compare those patterns, find agreements/disagreements and identify gaps.

Phase 9: Use those findings to create something academically useful and original: taxonomy, maturity model, evaluation hierarchy, conceptual framework, architecture and research agenda.

Phase 10: Put everything into a professional SLR report with PRISMA, tables, figures, critical discussion, references, appendices, Excel and slides.

Think of it like building a house: Phase 7 = understand the materials; Phase 8 = decide what the materials tell you; Phase 9 = design your own useful structure from the evidence; Phase 10 = build and present the final house.

PHASE 7 — ANALYSIS

Your question in Phase 7 is: “**What does our collection of included studies look like?**” Use the Phase 6 extraction Excel as your evidence base. The assignment lists publication details, AI technique, modelling stage, languages, validation, quality, scalability, reproducibility, industrial evidence and maturity among the extraction fields.

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1. Publication trends

Count studies by year, venue, country/institution, domain and contribution type. Make a table/graph. Then explain what the pattern means.

2. AI technologies

Count and compare LLMs, ML, DL, multimodal models, RAG, knowledge graphs, reinforcement learning, neuro-symbolic AI and agents. Do not only count: explain what each technology was used for.

3. Model-engineering stages

Map papers to requirements understanding, model generation/completion, validation/repair, transformation, model-to-code, code-to-model, synchronisation, evolution, runtime and adaptation. Look for over-researched and under-researched stages.

4. Modelling languages

Count UML, SysML, BPMN, DSLs, metamodels and other artefacts. Check whether research focuses on one diagram or supports multiple views.

5. AI roles

Identify whether AI is a requirement interpreter, generator/completer, recommender, critic/validator, repair assistant, transformation synthesiser, code generator, consistency monitor or autonomous orchestrator.

6. Validation mechanisms

Record human review, syntax/metamodel checks, OCL/formal constraints, consistency checking, simulation/model checking, execution/testing, traceability and hybrid AI+deterministic validation. ■filecite■turn0file1■L145-L148■

7. Quality

Compare correctness, completeness, consistency, semantic fidelity, traceability, maintainability, standards conformance, transformation accuracy, generated-code quality, reliability and explainability. ■filecite■turn0file1■L149-L151■

8. Industrial readiness

Ask: Was it tested only on toy/textbook examples, on realistic benchmarks, with industry partners, or in real/deployed systems? Also check scalability, cost, reproducibility and practical limitations.

9. Maturity levels

Place approaches on the assignment's continuum: L1 Manual, L2 Assistive, L3 Interactive co-modelling, L4 Semi-autonomous, L5 Verifier-guided autonomous, L6 Continuous adaptive modelling. Justify every assignment using evidence. ■filecite■turn0file1■L152-L163■

FULL-MARKS RULE: Never stop at “X papers used LLMs.” Add *so what?* Compare the studies, their evidence, validation, limitations and context. The rubric specifically rewards descriptive + thematic + comparative synthesis and critical reasoning. ■filecite■turn0file0■L141-L160■

PHASE 8 — SYNTHESIS

Your question in Phase 8 is: “**When we compare all the evidence, what do we learn?**” This is where your project becomes a real SLR rather than a list of papers.

A. Compare studies

Compare AI-generated vs human-generated models; AI vs rule-based transformations; LLM-only vs verifier-guided; prompting vs fine-tuning; proprietary vs open models; single-agent vs multi-agent; textbook examples vs industrial projects. These comparisons are explicitly suggested by the assignment. ■filecite■turn0file1■L264-L267■

B. Identify patterns

Look for repeated findings. Example structure: “Across X studies, researchers commonly used ____, while only Y studies reported ____.” Then explain why that matters.

C. Identify contradictions

Find cases where studies disagree. Ask whether the disagreement could be caused by different datasets, tasks, models, baselines, metrics or validation methods. Never hide contradictions.

D. Identify strengths and weaknesses

For each major approach, say what it does well and where it fails. Consider technical quality, validation, reproducibility, scalability and industrial evidence.

E. Build the gap matrix

Create columns such as: Research area | What studies have covered | Evidence available | What is missing | Why the gap matters | Future research needed. The assignment specifically asks you to identify underexplored areas and create a gap matrix. ■filecite■turn0file1■L304-L315■

FULL-MARKS RULE: A gap must come from your evidence. Do not write “more research is needed” without explaining exactly what is missing. For example: “Most studies evaluate syntactic validity, but few report semantic fidelity and cross-view consistency on realistic systems.” That is much stronger if your extraction data supports it.

PHASE 9 — SCHOLARLY CONTRIBUTION

Your question in Phase 9 is: **“What useful academic contribution can WE create from the evidence?”** This is the most creative part, but it must still be grounded in your SLR findings. The rubric expects original, logically derived and evidence-grounded artefacts. ■filecite■turn0file0■L162-L184■

1. Taxonomy

Create a structured classification of AI-DME. A strong structure can combine dimensions already provided in the assignment: lifecycle stage, input artefact, output artefact, AI technology, AI role, validation mechanism and quality attribute. ■filecite■turn0file1■L128-L151■

2. Maturity model

Show how AI-DME moves from little/no AI support to more autonomous operation. Use the assignment's levels as the evidence base, then explain what characteristics distinguish each level. Do not just copy the levels: explain what your reviewed studies show at each level.

3. Evaluation hierarchy

Create a hierarchy for judging AI-DME quality/trustworthiness. A useful evidence-based logic is to distinguish whether an output is syntactically valid, semantically faithful, complete, consistent, traceable, independently validated and practically/industrially useful. The exact hierarchy should be derived from your reviewed evidence.

4. Conceptual framework

Draw the main concepts and relationships. For example: AI technology → AI role → model-engineering task → model artefact → validation → quality/evidence → maturity/industrial readiness. Every relationship should be explained using your findings.

5. Reference architecture

Design a verifier-guided architecture showing AI, modelling tools/standards, deterministic verification, human oversight and traceability. The assignment specifically calls for a verifier-guided, standards-compliant, traceable and human-accountable architecture. ■filecite■turn0file0■L185-L205■

6. Research agenda

Turn the gaps into specific future research priorities. Rank them by importance/evidence. Avoid vague statements such as “future researchers should study AI more.” Say exactly what needs to be developed, tested or benchmarked.

FULL-MARKS RULE: Your artefacts must clearly come from the review. Add a small “Evidence basis” paragraph/table under each artefact explaining which analysis findings and studies led to each category, level or relationship. This makes your contribution traceable instead of looking copied or invented.

PHASE 10 — FINAL REPORT

Your question in Phase 10 is: **“Can a reader see exactly how we went from searching papers to our final conclusions?”**

1. PRISMA flow diagram

Show the numbers from identification/search → screening → eligibility/full-text → included studies. Include exclusions and reasons where required. The assignment says to follow PRISMA strictly and include the flow diagram. ■filecite■turn0file1■L304-L313■

2. Tables

Use tables for study characteristics, analysis results, comparisons, quality assessment, validation, gap matrix and scholarly artefacts. Tables should help the reader compare many studies quickly.

3. Figures

Use clear, numbered figures: publication trends, lifecycle distribution, technology distribution, validation landscape, maturity model, taxonomy, conceptual framework and reference architecture where appropriate.

4. Critical discussion

This is where you answer: What do the results mean? Why are some approaches stronger? What evidence is weak? What contradictions exist? What are the implications for trustworthy AI-DME? Do not simply repeat your results.

5. Conclusions

Answer the research questions using evidence. State the main findings, limitations and contribution. Do not introduce completely new evidence in the conclusion.

6. APA 7 references

Every claim based on a study needs appropriate citation. Make sure every in-text citation appears in the reference list and vice versa.

7. Appendices

Include supporting material such as search strings, inclusion/exclusion criteria, QA checklist/results, extraction/coding information, additional tables and other traceability evidence.

8. Excel extraction file

Submit the completed extraction/coding file. It must match the results reported in the paper.

9. Presentation/slides

Present the research question, method, PRISMA, major findings, critical gaps, scholarly contribution and conclusion. Keep slides visual and use tables/diagrams rather than huge paragraphs.

HOW 7 MEMBERS SHOULD SPLIT PHASES 7–10

Do **not** split it so that one person disappears after Phase 7. Each member should own a major evidence area from Phase 7 and carry that evidence into Phase 8 and Phase 9. Phase 10 is then assembled together.

Member	Phase 7 owner	Phase 8 job	Phase 9/10 job
1	Publication trends + coordinator	Connect trends to overall research landscape	Help write Results/Discussion; coordinate final report and PRISMA
2	AI technologies	Compare technologies and use cases	Contribute technology branch of taxonomy + discussion
3	Lifecycle stages	Find over/under-researched stages	Contribute lifecycle taxonomy + research gaps
4	Modelling languages + AI roles	Compare artefacts and AI responsibilities	Contribute taxonomy + conceptual framework
5	Validation mechanisms	Compare how outputs are checked	Lead evaluation hierarchy + verifier-guided architecture
6	Quality	Compare metrics, baselines, evidence strength	Lead quality/evidence discussion + maturity evidence
7	Industrial readiness + maturity	Compare real-world evidence and maturity	Lead maturity model + research agenda

Coordinator's job: keep one master coding dictionary, prevent double-counting, check that every graph/table matches Excel, and make sure every conclusion can be traced back to studies.

THE “FULL MARKS” FORMULA

- **1. Evidence:** Every important statement comes from your included studies/extraction data.
- **2. Comparison:** Compare approaches instead of describing papers one by one.
- **3. Critical thinking:** Ask why results differ, how strong the evidence is, and what limitations exist.
- **4. Traceability:** A reader should be able to move from a finding → table/figure → included studies → extraction data.
- **5. Synthesis:** Combine descriptive, thematic and comparative findings.
- **6. Gaps:** Identify precise under-researched areas from the evidence.
- **7. Contribution:** Build your taxonomy, maturity model, evaluation hierarchy, framework, architecture and research agenda from those findings.
- **8. Presentation:** Clear academic writing, numbered tables/figures, APA 7, PRISMA and professional formatting.
- **9. Consistency:** The numbers in Excel, report, graphs, tables and presentation must agree.
- **10. No unsupported claims:** If a paper does not report something, say “not reported”; do not guess.

FINAL CHECK BEFORE SUBMISSION

- Phase 7 has all 9 analysis areas.
- Phase 7 contains counts AND interpretation, not only counts.
- Phase 8 contains comparisons, patterns, contradictions, strengths/weaknesses and a detailed gap matrix.
- Phase 9 contains evidence-based taxonomy, maturity model, evaluation hierarchy, conceptual framework, reference architecture and research agenda.
- Each Phase 9 artefact explains how it was derived from the SLR evidence.
- Phase 10 contains PRISMA, tables, figures, critical discussion, conclusions, APA 7 references, appendices, Excel and slides.
- Every important finding can be traced to included studies.
- Quality-assessment scores are used to weight/interpret evidence rather than ignored. The assignment says QA scores should inform evidence weighting. ■filecite■turn0file1■L212-L213■
- The final report does not merely summarise studies.
- The group has checked the entire report for consistent numbers, terminology, citations and formatting.

ONE SENTENCE TO REMEMBER

Phase 7 tells us WHAT the research looks like → Phase 8 tells us WHAT IT MEANS → Phase 9 tells us WHAT WE CAN CONTRIBUTE → Phase 10 proves and presents the whole journey.

The rubric's strongest expectations are exactly these: rigorous methodology, evidence synthesis and critical analysis, original evidence-based scholarly artefacts, a traceable contribution, and professional academic presentation.

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